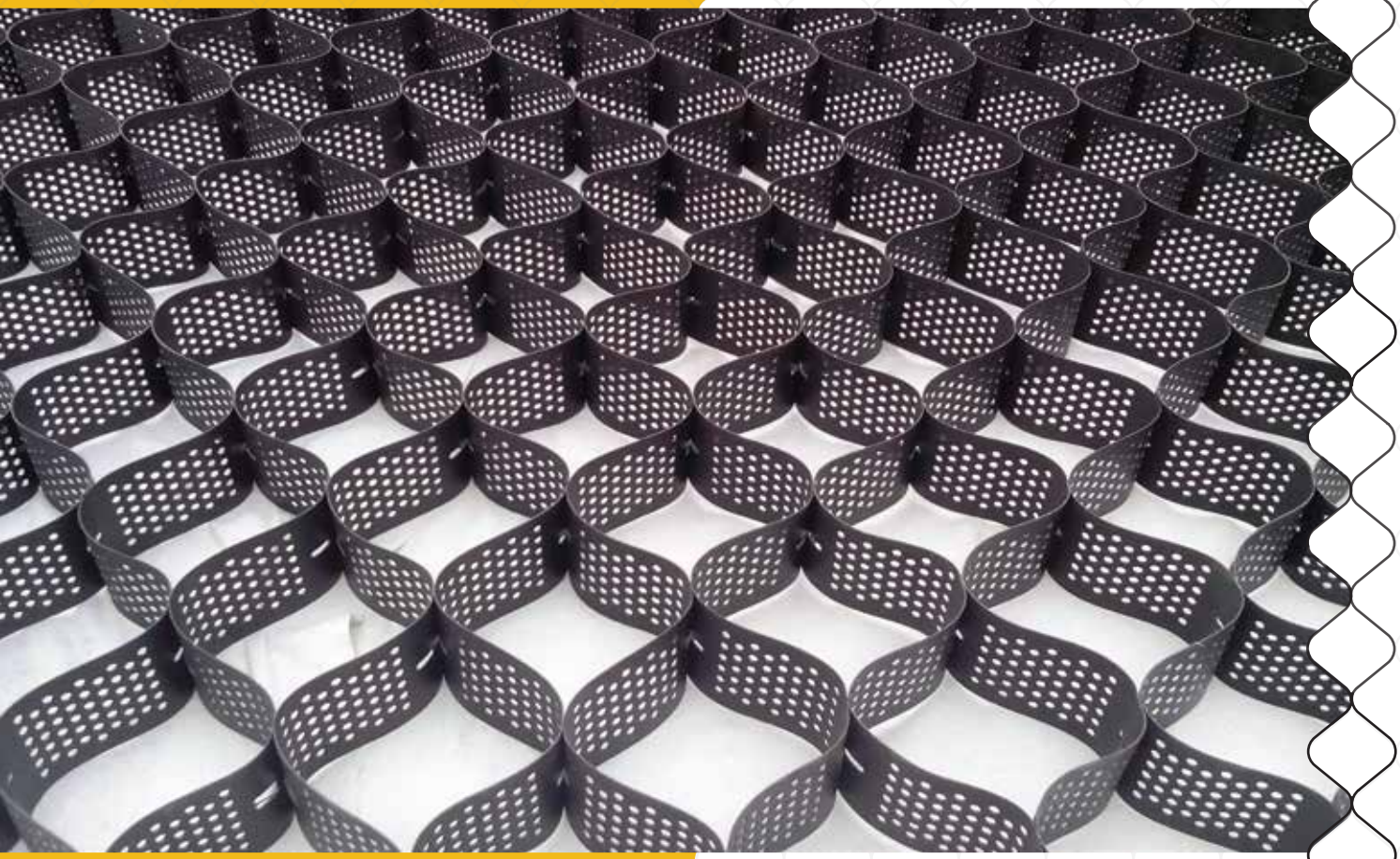
LIGHT
STRONG
FAST



Geocell

CELLULAR CONFINEMENT SYSTEM

What is Geocell ?



LSF
LIGHT
STRONG
FAST

Cellular Confinement System

GEOCELL

www.lsf-egypt.com

Geocell (Cellular Confinement System) is a high quality erosion control and soil stabilization material with a three-dimensional honeycomb structure. The system provides fast, economical and long lasting solutions for soil stabilization applications and hydraulic structures, while preventing earth loss due to water and wind in general purpose sloping surfaces.

Geocell, produced by frist quality polymer strips with special additives with innovative ultrasonic welding technology, provides excellent protection and reinforcement on the applied surface.

While the perforated structure of the material allows fast drainage of water, the textured surface of the cells allow the filling material to be reliably held even at high slopes, Cells Can be filled with different materials (sand, gravel, soil, concrete) according to the purpose of use.

Cell heights vary between 300-50 mm depending on project requirements.

Geocell Applications

Erosion Control



Roads



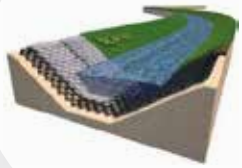
Landscape



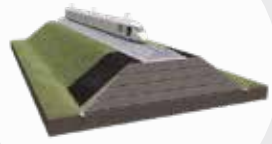
Retaining Walls



Hydraulic Structures



Railways

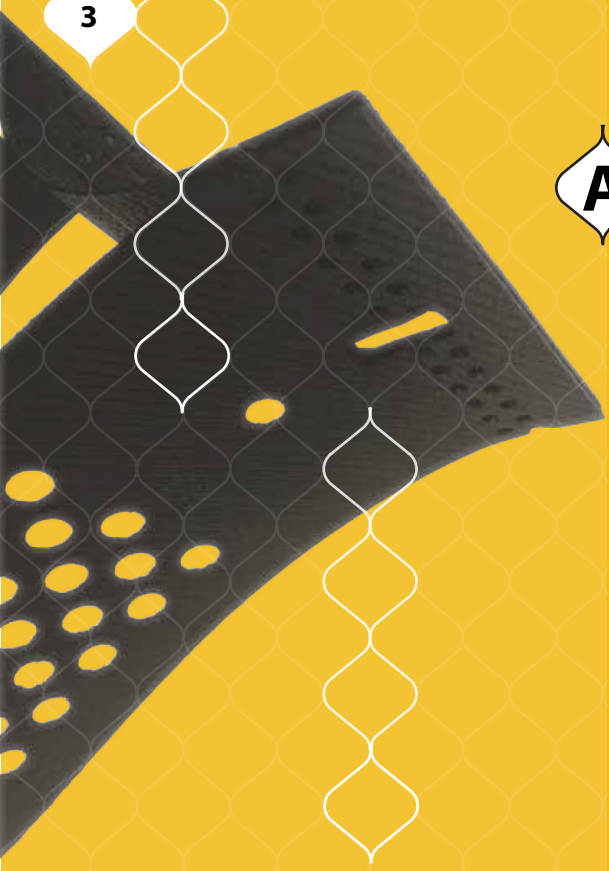


Airports



Port Container





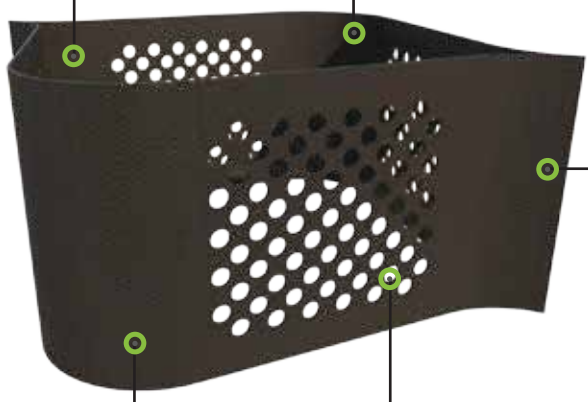
PERFORATED - TEXTURED GEOCELL WALL

ADVANTAGES

- Perforations facilitate parallel slope drainage of the in filled cell. In saturated conditions, the removal of excess water increases infillfriction, reducing down slope sliding forces, resulting in a more stable system.
- In vegetated systems, perforations allow roots to grow from cell-to-cell creating greater vegetative stability against short-term hydraulic events. Perforations also allow nutrients and microorganisms to pass through neighboring cells.
- Textured surfaced cells significantly increase the frictional resistance of the cell wall with the infill material. Frictional surface limits the movements and the loss of the infill materials due to freeze-thaw cycles, water and other eroding factors.

Why LSF Geocell ?

High Quality Materials
Geocell is manufactured from the materials. Cells reinforced highest quality with specially added polymers serve safely even in the most difficult natural conditions.



Increased Strip Thickness
Geocell has superior strength against tensile loads with strip thickness of 1.20 millimeters and over.

Excellent Welding Strength
Geocell has a special ultrasonic welding system with superior strength that does not break even under extreme loads.

Textured Cell Wall
The specially designed textured frictional surface prevents the loss of material in the cell by increasing the and the fill material.

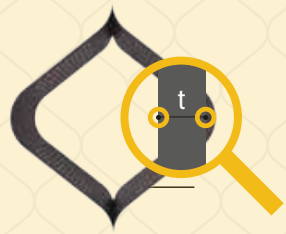
Perforated Cell Wall
The increased perforated structure permits rapid drainage of water between cells and propagation of the plant root system.

Technical Data









W WELDING DISTANCE

The distance between two welding points of the cell. The cell size increases as welding distance increases.

AxB CELL SIZE

(A) is the transverse and (B) is the longitudinal inner distances of the opened cell.

H CELL HEIGHT

It is determined by the height of the material to be filled into the cell according the slope increases

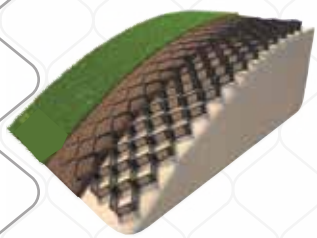
t STRIP THICKNESS

It is the strip thickness of the cell walls. Increasing the tensile strength with more strips thickness.

Welding Distance (W)	Cell Width (A)		Cell Length (B)	Cell Height (H)	Strip Thickness (t)
330 mm	210 mm	X	245 mm	300/200/150/100/75/50 mm	2,0 - 1,2 mm
400 mm	260 mm	X	290 mm	300/200/150/100/75/50 mm	2,0 - 1,2 mm
450 mm	280 mm	X	340 mm	300/200/150/100/75/50 mm	2,0 - 1,2 mm
500 mm	340 mm	X	365 mm	300/200/150/100/75/50 mm	2,0 - 1,2 mm
600 mm	390 mm	X	420 mm	300/200/150/100/75/50 mm	2,0 - 1,2 mm
660 mm	420 mm	X	490 mm	300/200/150/100/75/50 mm	2,0 - 1,2 mm
700 mm	448 mm	X	520 mm	300/200/150/100/75/50 mm	2,0 - 1,2 mm

Infill Materials

Erosion Control



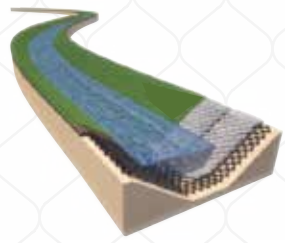
» Vegetative Soil
» Vegetative Soil & Crushed Stone
» Crushed Stone

Soil Stabilization & Retaining Walls



» Mechanical Fill
» Crushed Stone

Hydraulic Structures



» Vegetative Soil
» Vegetative Soil & Crushed Stone
» Crushed Stone
» Concrete

Supplies

- Anchorage Bar
- Plastic Clips/Stapling
- Geotextile
- Tendon
- Geogrid

Hand Tools

- Hammer
- Driller
- Shovel
- Compactor



EROSION CONTROL

INFILL MATERIALS



Vegetative Soil



Crushed Stone



Vegetative Soil & Crushed Stone

cellular confinement system protects a number of engineering structures, such as high containment dikes and levees, abutments against erosion and landslides. way and railway side slopes, Even on extremely sloped surfaces, the system prevents the valuable material on the slope surface from being lost due to environmental factors (rain, water, wind etc.)

The cell size and cell height are determined according to the slope height and slope to be applied. the type of infill material to be used has also an important influence on cell size selection.

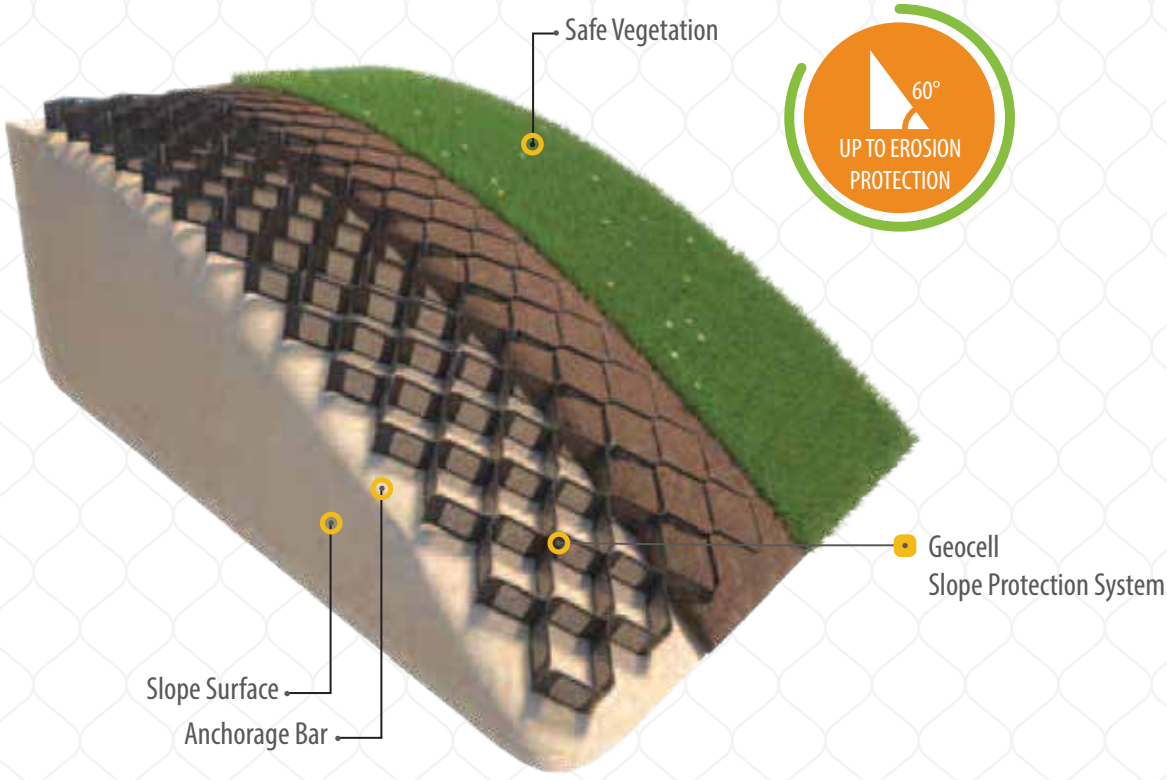
Weathering erosion is the removal of valuable vegetative soil by water and the transport of soil particles through small cracks. As abrasion and decomposition increase, or as the flow rate of water increases, the gutters deepen and expand.

Geocell significantly increase the performance of slopes against erosion by controlling the flow of water and the migration of fill material inside the cells.

EROSION CONTROL

»Cut or Fill Embankment Slopes
»Shoreline Revetments
»Containment Dikes and Levees
»Landscape Applications
»Landfill Linings and Covers
»Abutment Protection
»Highway and Railway Side Slopes
»Dams and Spillways
»Vegetated Channel Structures





WEATHERING EROSION





ROOT PROTECTION [VEGETATIVE SOIL]

Reinforces root systems and directs hydraulic flows over the top of cells, with the cells acting as a series of check dams.



GRANULAR PROTECTION[CRUSHED STONE]

Improves the performance of granular filled slopes by controlling the migration of fills that would beinitiated by hydraulic and gravitational forces. otherwise

Advantages

- Allows aesthetic landscape applications.
- Prevents the loss of material on the slope surface.
- Increases slope stability by regulating water intake from the surface.
- Protects the root system of plants and provides safe vegetation.



In slopes suitable for cultivation, the cells are filled with agricultural soilColorful plant seeds can be placed to provide an aesthetic, attractive appearance



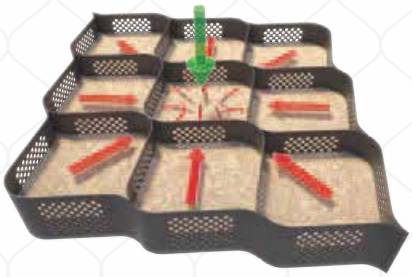


In slopes that are not suitable for planting, the cells are filled with colored stones to provide a permeable and aesthetic appearance.



SOIL STABILIZATION

CELLULAR CONFINEMENT



Horizontal movements of the infill material are prevented by the confinement effect of the interconnected three-dimensional cells. system works with a flexible base principle and The spreads the loads that affect the system to a wider area.

Geocell forms a three dimensional load support system that significantly increases the load bearing capacity of the underlying subgrade soil.

Each cell confines the soil, preventing lateral spreading and increasing base stiffness. Both soil interaction with the cell walls and passive resistance developed between adjoining cells increases vertical shear resistance of the infill material. When filled with granular infill, Geocell can reduce base course thickness by as much as %50.





SOIL STABILIZATION

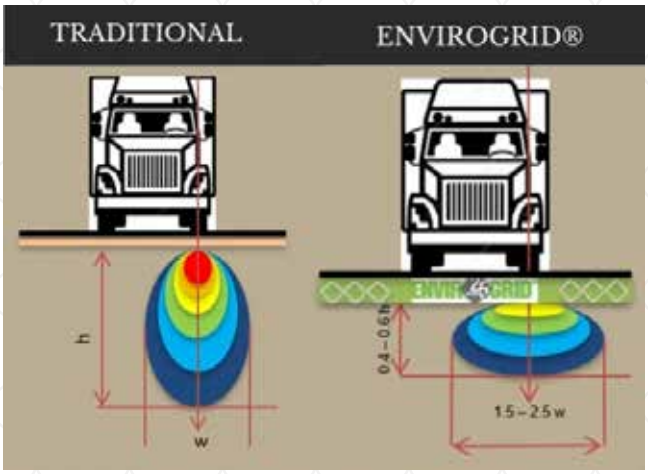


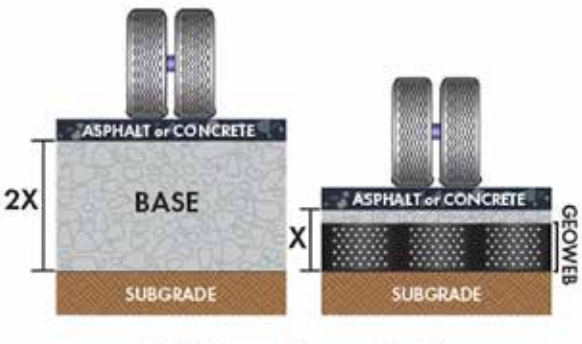
» Stabilized Drainage Layer
» Permanent and Temporary Access Roads
» Permeable Load-Support
» Roadway Shoulders
» Stabilized Base For Asphalt Pavements
» Track Ballast and Subballast Structures
» Trails and Walkways
» Foundation Mattresses and Pipeline Protection
» Sports Fields and Playgrounds
» Parking Areas

Advantages

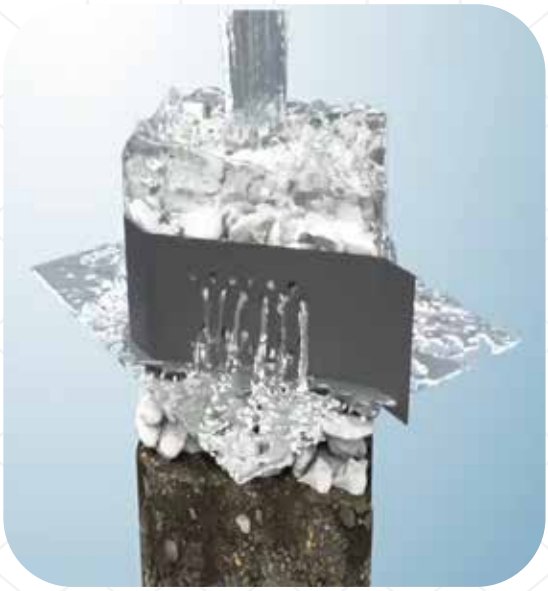
- Significantly minimizes surface rutting.
- Increases load bearing capacity of the soil.
- Allows fast and controlled soil compaction.
- controls shearing and lateral movement of the coarse infill material.
- Distributes loads laterally and reduces vertical deflection.

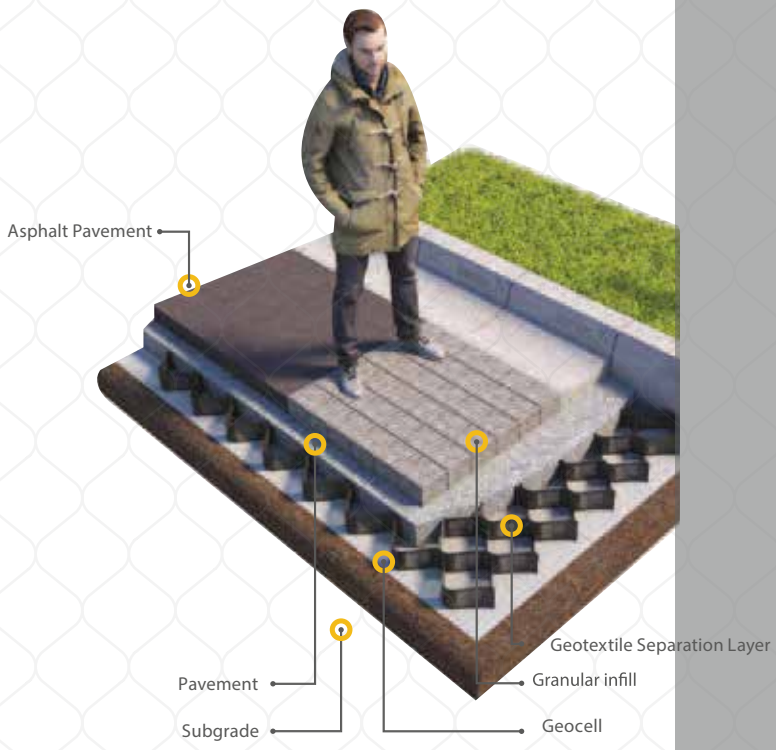






50% Less Cross Section




Asphalt Pavement
Pavement
Subgrade
Geotextile Separation Layer
Granular infill
Geocell



STABILIZED SUPPORT SYSTEM

With the use of Geocell, vehicle and pedestrian loads are distributed over a larger area instead of a direct contact point.



Stabilized 3D support system.

Thanks to the superior confinemet effect of the system, the load concentration on the base decreases singnificantly. Due to the high drainage capacity of the material, settlements are reduced dramatically.



SPORTS FIELDS
PLAYGROUNDS



BICYCLE/MOTORCYCLE
TRAILS AND WALKWAYS



SAND STABILIZED
ACCESS ROADS

> In areas with limited easements, Flexi Geocell retaining wall can be constructed by filling the cells with concreteor crushed stone.



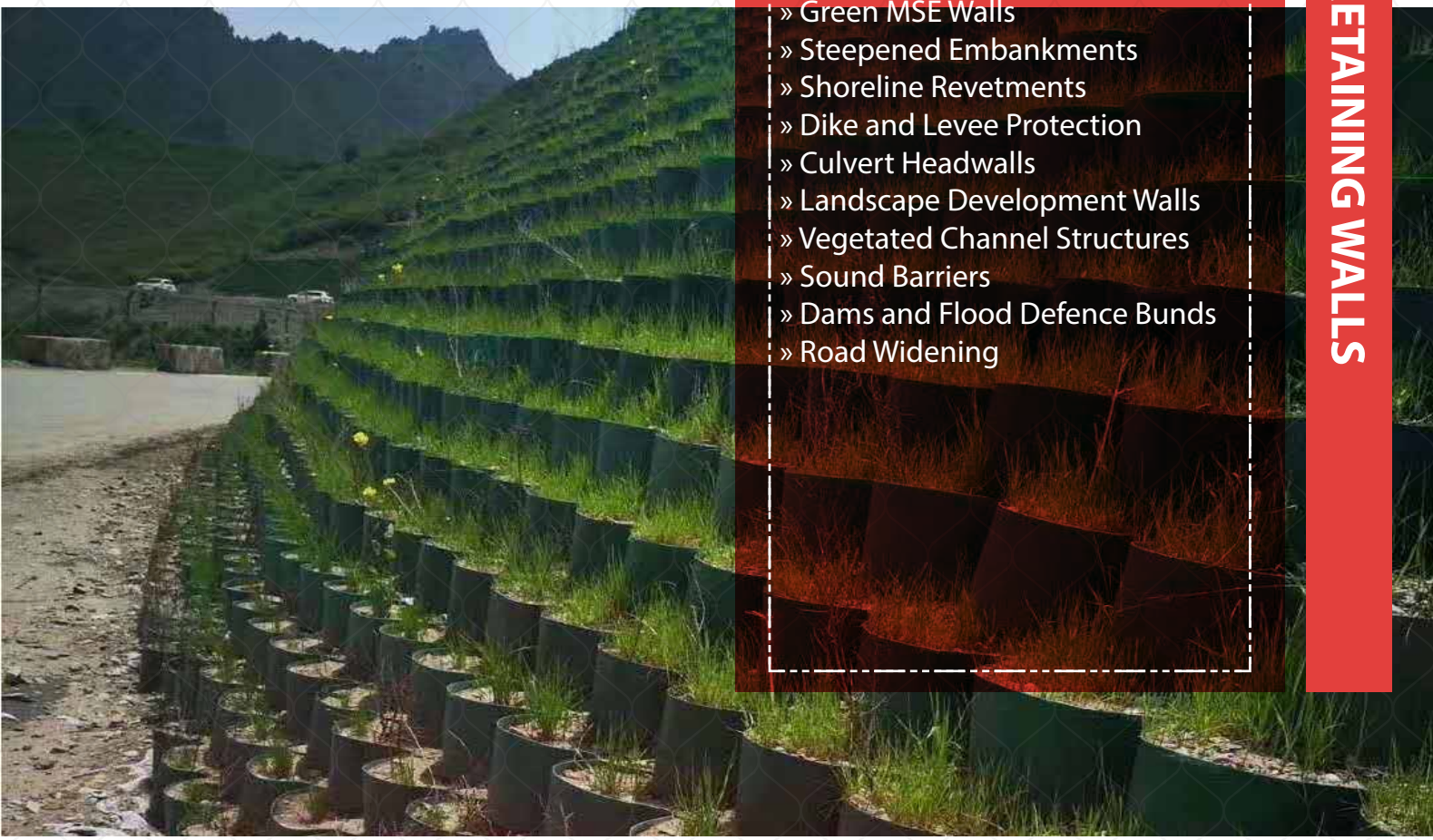
RETAINING WALLS

Flexi Geocell retaining walls manufactured using the Cellular confinement System are green retaining Structures that do not require concrete and mold, are perfectly compatible with nature and are extremely resistant to earthquake.

Flexi Geocell walls can easily reach the height that conventional retaining walls (stone, reinforced concrete) can not reach. The system allows for construction flexibility and provides aesthetics through a completely vegetated face.

- It has an aesthetic vegetated front face integrated with nature.
- Fits with horizontal and vertical geometry perfectly.
- Retaining walls with unlimited height can be manufactured.
- Its flexible structure provides excellent performance against earthquake.
- The system does not require concrete and mold.

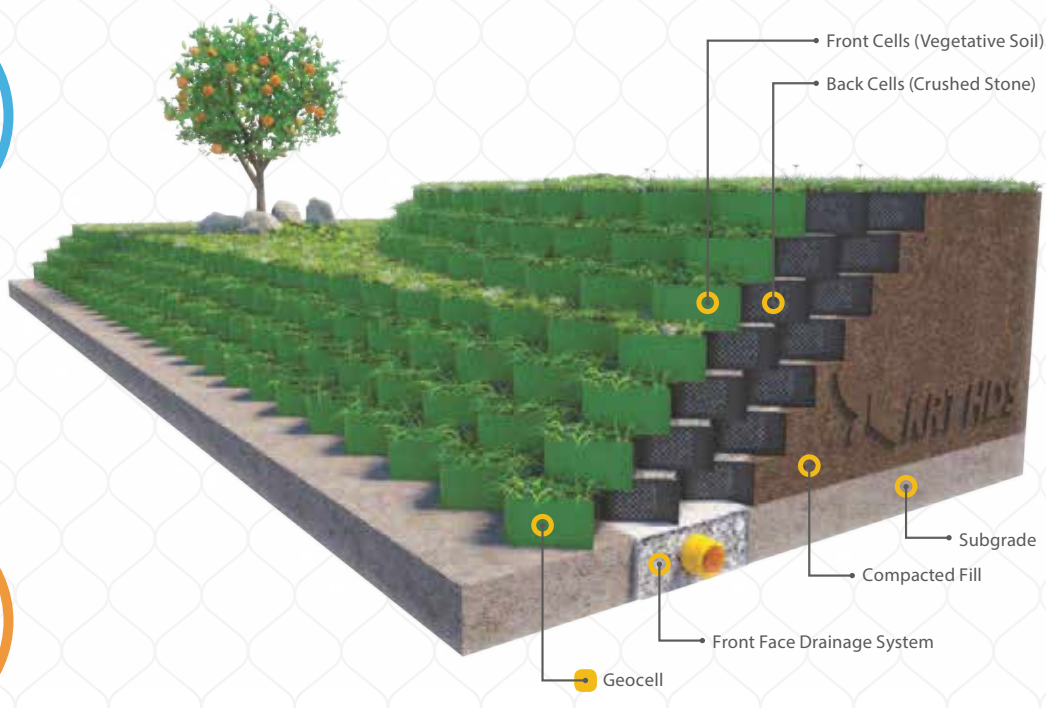
RETAINING WALLS



- » Green MSE Walls
- » Steepened Embankments
- » Shoreline Revetments
- » Dike and Levee Protection
- » Culvert Headwalls
- » Landscape Development Walls
- » Vegetated Channel Structures
- » Sound Barriers
- » Dams and Flood Defence Bunds
- » Road Widening

PERFECT PERFORMANCE AGAINST EARTHQUAKE

UNLIMITED WALL HEIGHT

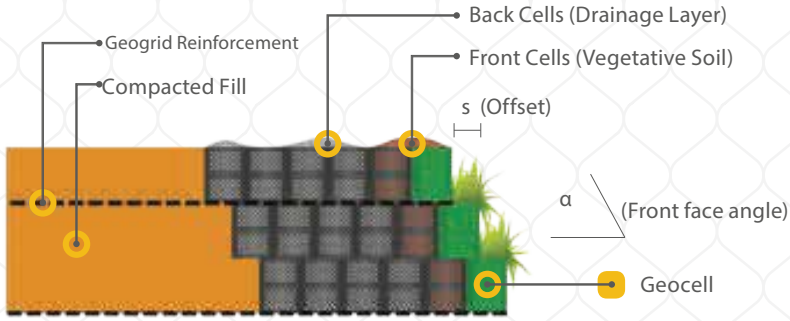


FLEXI WALL FRONT FACE



In Flexi Geocell retaining wall applications, the cells behind the cells filled with vegetable soil on the front of the wall are filled with a crushed stone so that the water can be transferred to the continuous drainage system at the base of the wall.

Flexi Geocell retaining walls constructed to ensure the stability of the steep slopes against the landslide can be manufactured with the method of laying geogrid reinforcements and compacting high quality filling material at every stage.Desired front face angle (a) of the wall can be achieved by setting offsets(s) at every stage with a prescribed distance. The main components of the wall system are Geocell wall element, high quality fill material and geogrid reinforcements.



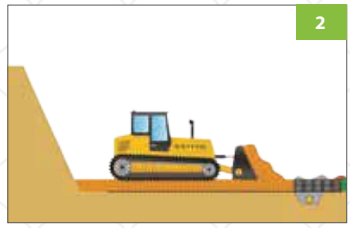
Flexi Geocell Retaining Wall Cross Section Detail



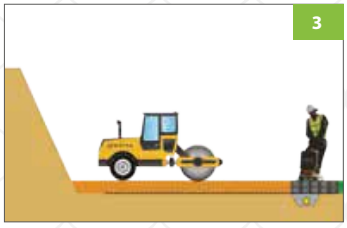




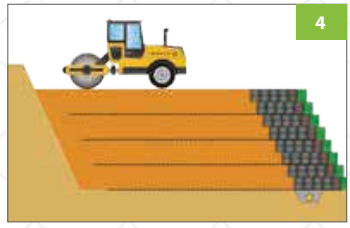
Levelling and drainage excavation are done according to the project. Geocell and other materials are installed.



At each level of the compaction laying of geogrid and geocell is done with extensive care so that the geogrids are not damaged.



The front face cells are filled with crushed stone and vegetative soil in each step and compacted with vibrating plate compactor.



In order to achieve desired front face angle of the wall, offsets are set with Geocell at each level of laying/compaction.



The application of the wall is completed when the laying and compacting process reaches the specified height Geocell front cells are planted to create aesthetic natural look.



GRANULAR DRAINAGE SYSTEM



Thanks to high drainage capacity of Geocell system with perforated cell walls, the flow of water is directed to the cell surface, preventing from erosion on the slope surface and on the channel bed.

Geocell offers a wide variety of flexible protection applications for hydraulic structures. the cellular confinement system provid and protection of channels exposed to erosive conditions ranging from esstability low-to-high flows either intermittent or continuous.

The system greatly improves the hydraulic performance of conventional protection materials such as aggregate, rip-rap and vegetation by confining them with in the cellular structure Geocell system can be environmental, ecological and aesthetic requirements.

- Can be used in Intermittent or continuous/low to high flow channels.
- Offers different infill alternatives depending on the flow conditions.
- Increases the performance of conventional protection materials.
- Reduced installation costs.



- » Stormwater Diversion or Containment
- » Flow Channels
- » Shore Protection
- » Culvert Outfalls
- » Dam Faces and Spillways
- » Down Chutes
- » Drop Structures
- » Swales and Drainage Ditches
- » Process Water Channels
- » Storm Water/Waste Water Lagoons
- » Geomembrane Protection

INFILL MATERIALS



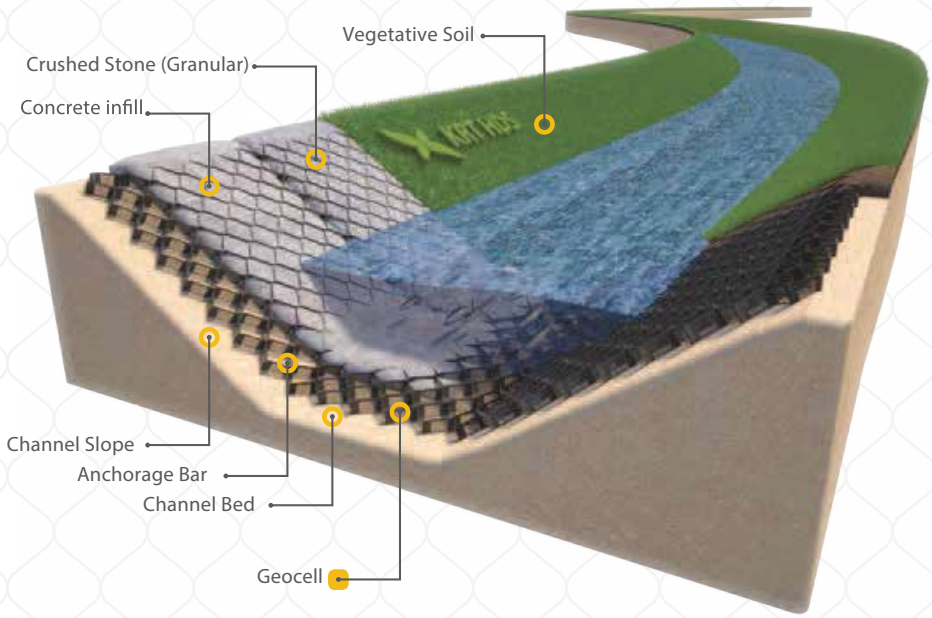
Vegetative Soil



Crushed Stone



Concrete







Geocell
Cellular Confinement System

Description	Value	Test method
Material composition	UV Resistant Polyethylene(HDPE)	
Height/ Cell Depth(H) mm	50-75-100-150-200-250-300 (±10%)	
Welding Distance(W) mm	330-400-660(+-10%)	
Nominal Expanded Cell Size mm	210X250 - 210X340 - 470mm=490	
Number of Cells /m2	5-15	
Color	Black	
Strip Surface Treatment	Textured Surface Perforated/ Non perforated	
Welding Type	Ultrasonic	
Strip Thickness	1.15 mm (COV < 2%)	ASTM D-5199
Material Density Kg/m³	1.33 (COV < 2%)	ASTM D-1505
Carbon Black Content	2.5 – 2.5 %	ASTM D-1603
Tensile Strength (Mean) kN/m	17 (COV < 5%)	EN ISO 10319
Seam Peel Strength (Mean) N/m	12000 (COV< 10%)	EN ISO 13426
Heat deflection temperature	64 -77 C°	
Texture Amplitude (Mean) (mm)	0.35	



Al- Azhar University
جامعة الأزهر

المشروع القومي لتأهيل وتبطين الترع
تقرير فني عن استخدام تقنية الجيوسيل في أعمال التبطين

مقدم الى
الهيئة العربية للتصنيع (مصنع قادر)

إشراف:
مركز الأعمال والإستشارات الهندسية - جامعة الأزهر

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أستاذ بكلية الهندسة جامعة الأزهر

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Page 1 of 15

FLOW RATE: 3,5-2,0 m/s



CONTINUOUS FLOW
INTERMITTENT FLOW

CRUSHED STONE INFILL
PERMEABLE PROTECTION

performs well, allowing the use of different sizes for variances in flow velocities encountered from site to site.

- Allow to use smaller diameter, less costly aggregate by confining and improving the performance of infill material.
- Cost effective alternative to large rip rap or hard armoring protection.
- Directs flow over the top cells: there by increasing the shear resistance of the fills.
- Limits the movement of the aggregates due to the eroding effect of flow.

FLOW RATE: 6,0-0,0 m/s



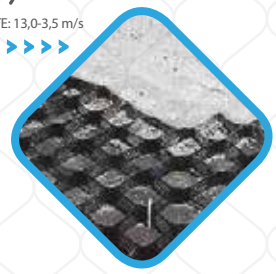
INTERMITTENT FLOW

VEGETATIVE SOIL
PLANT ROOT PROTECTION

Provides an aesthetically pleasing look and performs exceptionally well in applications with low to moderate flows.

- Reinforces root zones and directs flow over the top cells; there by increasing the shear resistance of the fill.
- Confined vegetative soil performs well where the flow is intermittent.
- Prevents formation of cracks.
- Compared to conventional methods, it offers a more aesthetically superior solution.

FLOW RATE: 13,0-3,5 m/s



CONTINUOUS FLOW
INTERMITTENT FLOW

CONCRETE INFILL
RIGID PROTECTION

A Cost effective alternative to traditional installed concrete lined channels.

- Installation costs are dramatically reduced through elimination of costly forms and other construction techniques.
- Increases the stability singnificantly in steep slopes where the flow rate is relatively high.
- Can produce a flexible concrete slab for a low-friction, armored channel lining.
- Permits conformance with subgrade movement without the potential cracking and undermining associated with poured-in-place concrete slabs.







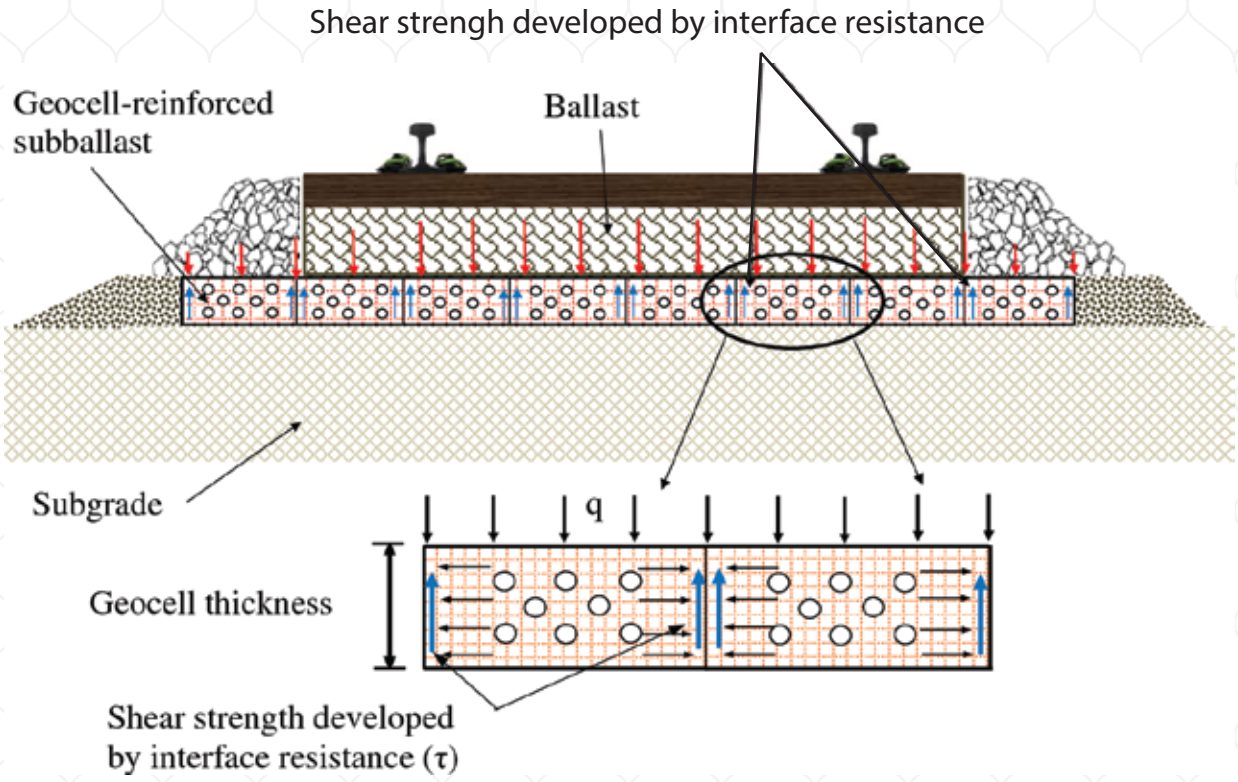
Geocell reduce maintenance, and have been demonstrating the most significant cost saving in railways construction and maintenance, by eliminating the need for high quality construction materials, and while improving the structure for long-term applications. The growing infrastructure activities across the globe are griving the geocell market as these composites offer high quality and are cost-effective alternative in various applications.





» Geocell technology is the best solution to increase saving cost efficiency in Railways Applications and maintenance contributes to the anticipated growth of this field.

» Geocell are used in all surfaces of the transportation industry in cluding roadways, airport, rail roads and waterways.



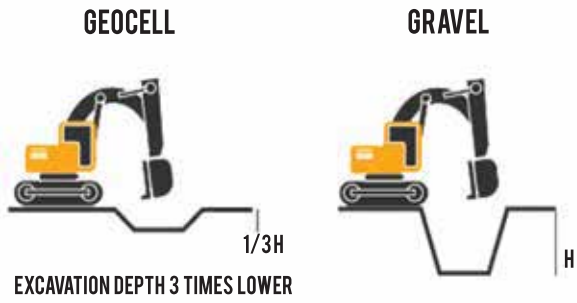


» Geocell is the best selection founded by authorities as they have been searching for innovative techniques special for airport road runways application, which offer a suitable reinforcement alternative to the conventional method, that improves the overall structureal strength, reduce operational costs, minimizing maintenance requirements and uses on-site or recycled materials.



Geocell have the increasing demand in Airport applications is accounted for the emergence of this solution, mainly as it offers an alternative to costly granular materials and to distant quarries stabilization improves the bearing capacity of soft soils. This increases the strengh of shoulders to safely support even heavy aircraft loads if required in an emergency situation.

EXCAVATION DEPTH



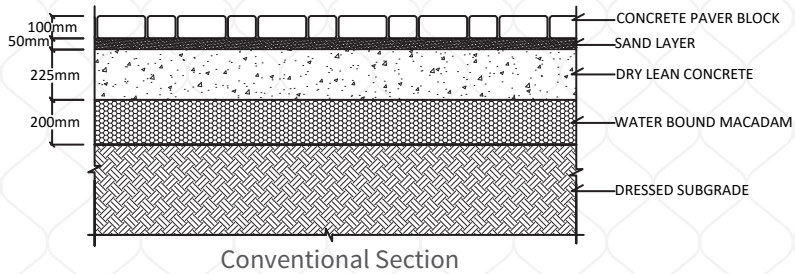
EXCAVATION DEPTH 3 TIMES LOWER

Container port

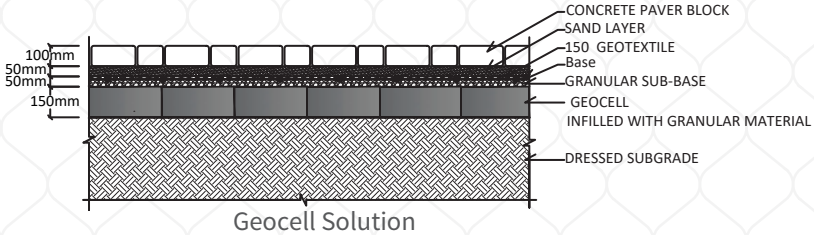


LSF - Egypt of geosynthetic experts proposed a solution with geocells and a non-woven geotextile. They recommended expelling a layer of thickness of the yard crust section and replacing it with a layer of geocells using GSB (Granular Sub-base) as infill, completely eliminating the expensive concrete section. Introducing Geocell in the granular base or sub-base improves the life span of the pavement and also economizes the overall crust thickness. Geocell increases the stiffness of aggregate base layer and confines the aggregate particles under repetitive loading.

Technical details



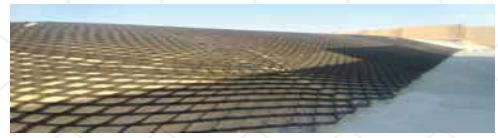
Conventional Section



Geocell Solution







Sharm El Sheikh project







Galala Project





Ismailia project









Ain Sokhna Project

HYDRAULIC STRUCTURES







Tanta project

HYDRAULIC STRUCTURES







Ismailia project

HYDRAULIC STRUCTURES



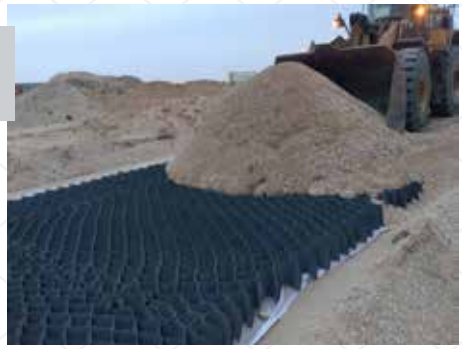




Banha project

Container port









Port Said project

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STRONG
FAST

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